Mohand Ouidir AMIRAT

3rd year PhD Student at Université Claude Bernard Lyon 1
Born on July 02, 1998 in Annaba, Algeria / Nationality: Algerian
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Research Interests

My research interests lie within the field of Control Theory for Partial Differential Equations (PDEs). I am particularly interested in the analysis and stabilization of nonlinear hyperbolic equations, with a specific focus on *weakly hyperbolic systems of conservation laws*. My work also involves numerical analysis and scientific computing, essentially for models arising from fluid mechanics and/or process engineering.

Education

PhD in Automatic Control (Specialization: Control Theory) *2023 - Oct 2026 (Expected)*
LAGEPP UMR CNRS 5007, Claude Bernard University Lyon 1 - Lyon, France

Thesis Title:	Boundary stabilization of 1-D weakly hyperbolic systems of conservation laws: application to sedimentation in wastewater treatment
Supervisors:	Vincent Andrieu, Mathieu Bajodek and Claire Valentin
Funding:	Doctoral fellowship from the French Ministry of Higher Education and Research, obtained through the EEA Doctoral School (Lyon)

M.Sc. in Mathematics - Modeling and Numerical Analysis (Full 2-year program) *2021 - 2023*
Université of Montpellier - Montpellier, France

1st year M.Sc. in Applied Mathematics / Rank: 2^{nd} in class *2020 - 2021*
Badji Mokhtar University - Annaba, Algeria

B.Sc. in Mathematics / Rank: 3^{nd} in class *2017 - 2020*
Badji Mokhtar University - Annaba, Algeria

High School Diploma / Distinction: Honors (Assez Bien) *2016 - 2017*
Specialization: Technical-Mathematics - Algeria

Publications

Articles in International Journals

- **M.O. Amirat**, M. Bajodek, V. Andrieu, J. Auriol, C. Valentin. *Insights into the stability analysis of 2-by-2 linear weakly hyperbolic systems.* Provisionally acccepted for publication in *Automatica*, 2026.
- J. Auriol, A. Braun, **M. O. Amirat**, M. Bajodek. Underactuated boundary control of a linearized Saint-Venant equation. *IMA Journal of Mathematical Control and Information*, 2025.
- *In preparation* – **M.O. Amirat**, V. Andrieu, M. Bajodek, C. Valentin. *Stabilization of a shock steady-state for weakly hyperbolic systems of balance laws: application to wastewater treatment.* 2026 (to be submitted).

Conference Proceedings

- **M.O. Amirat**, V. Andrieu, J. Auriol, M. Bajodek, C. Valentin. *Boundary feedback control of a 2×2 weakly hyperbolic system: Lyapunov-based approach. IFAC-PapersOnLine, Vol. 59, Issue 8,* 2025
- **M.O. Amirat**, V. Andrieu, M. Bajodek, J. Auriol, C. Valentin. *Stability analysis of a class of 2×2 triangular hyperbolic systems. IFAC-PapersOnLine, Vol. 58, Issue 21,* 2024.

Talks & Communications

International Conferences

- 5th IFAC/IEEE-CSS Conference on Control of Systems Governed by Partial differential equations (CPDE 2025), Beijing, China. *June 2025*
- 4th IFAC Conference on Modelling, Identification and Control of Nonlinear Systems (MICNON 2024), Lyon, France. *Sept. 2024*

National Conferences

- Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées (CJC-MA 2026), ENPC, Champs-sur-Marne, France. *March 2026*
- 2^{eme} Congrès annuel de la société d'Automatique, de Génie Industriel et de Productique (SAGIP). Villeurbanne, France. *May 2024*

Seminars & Workshops

- Spring School on Theory and Applications of Port-Hamiltonian Systems (PHS 2025), Fraueninsel, Germany. *March 2025*
- DYCOP Team Seminar, LAGEPP. Université Claude Bernard Lyon 1, France. (*2024, 2025 and 2026*)

Teaching Experience

Teaching assistant at the Department of Electrical and Process Engineering — *2023 – Present*
Claude Bernard University Lyon 1
Total: 144 teaching hours

Numerical Approximation of PDEs (2nd year graduate students in Control Engineering)
Design and teaching of the last part of the course "Distributed Parameter Systems" (numerical part).

- Finite Difference schemes for Heat and Advection-Diffusion equations.
- Finite Volume methods for nonlinear hyperbolic systems (1-D Saint-Venant).
- Godunov schemes and approximate Riemann solvers. Development of computational codes.

Numerical Methods for Engineers (1st year Graduate students in Control Engineering)
Direct and Iterative methods for Sparse Linear Systems and numerical optimization (Gradient descent, Conjugate gradient methods, and least-squares problems).

Advanced Programming: C Language (3rd year undergraduate students in Electrical Engineering)
Pointers and arrays, memory allocation, modular programming (sources, headers, makefiles), and low-level programming (bitwise operators, masks, bit-fields).

Introduction to Automatic Control (2nd year undergraduate students in Electrical Engineering)
Laplace Transform and Convolution, Stability analysis, Frequency domain, Feedback control loops, Elementary controllers (P, PI, PID).

Responsibilities

Co-organizer of the PhD Seminar (Café Contrôle) — *2025 - Pres.*
LAGEPP UMR CNRS 5007, Claude Bernard University Lyon 1 - Lyon, France
Mission: Speaker selection, logistical management, and session moderation to foster scientific exchange among PhD students and Post-Doc researchers in the laboratory.

Research Experience & Master Internships

2nd year M.Sc. Internship: Development of a Godunov-type Finite Volume scheme for the numerical approximation of a weakly hyperbolic system — *March – Aug 2023*
Application to sedimentation in wastewater treatment
LAGEPP UMR CNRS 5007, Claude Bernard University Lyon 1 - Lyon, France
Supervision: Frédéric Lagoutière, Claire Valentin and Christian Jallut

Abstract: Implementation of a Finite Volume solver for nonlinear hyperbolic systems. Development of splitting strategies. Comparison with experimental data and parameter estimation through optimization.

1st year M.Sc. Internship: Numerical methods for Hamilton-Jacobi equations and viscosity solutions
March – June 2022
IMAG, University of Montpellier - Montpellier, France
Supervision: Jessica Guerand

Abstract: Theoretical and numerical study of Hamilton-Jacobi equations and viscosity solutions. Extension to the non-stationary case, including the analysis and implementation of a Finite Difference scheme with optimal error estimates.

Technical Skills

Languages: C/C++, MATLAB, Python, FreeFEM++, FEniCS, HTML/CSS

Software & Tools: ParaView, Gnuplot, Gmsh, LaTeX, Git/GitHub, VS Code

Operating Systems: macOS, Linux (Ubuntu), Windows

Languages

French: Native/Bilingual (C2) | **English:** Advanced (C1)
Berber: Mother language | **Arabic:** Native/Bilingual (C2)